

Quang Hieu Pham

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Personal Statement

My research explores two core questions: how AI models can more accurately reflect human cognition and what characterizes their reasoning processes. I am particularly interested in the “Era of Experience” where learning occurs through interactions with the real world.

Education

University of Alberta (UofA)

Sep 2025 – Ongoing

M.Sc. in Computing Science (Thesis-based)

- GPA: 4.0/4.0
- Focuses: Agentic, Reasoning, Reinforcement Learning, Text-to-SQL

Hanoi University of Science and Technology (HUST)

Aug 2019 – Aug 2023

B.S. in Data Science

- Graduation GPA: 3.88/4.00
- Top (2/56) within DS major, top 7 within School of Information Technology.
- Thesis: [Abnormal Event Recognition from Videos by Prompting Vision-Language models](#).

Uppsala University (UU)

Aug 2021 – Jan 2022

Exchange Student at Department of Information Technology

Publications & Preprints

[*] denotes equal contribution. Reverse-chronological order.

[FlexSQL: Flexible Exploration and Execution Make Better Text-to-SQL Agents](#)

Preprint 2026

Quang Hieu Pham, Yang He, Ping Nie, Canwen Xu, Davood Rafiei, Yuepeng Wang, Xi Ye, Jocelyn Qiaochu Chen

[Detecting and Suppressing Reward Hacking with Gradient Fingerprints](#)

Preprint 2026

Songtao Wang, **Quang Hieu Pham**, Fangcong Yin, Xinpeng Wang, Jocelyn Qiaochu Chen, Greg Durrett, Xi Ye

[ClozeMath: Improving Mathematical Reasoning in Language Models by Learning to Fill Equations](#)

ACL 2025 (Findings)

Quang Hieu Pham^{*}, Thuy Duong Nguyen^{*}, Tung Pham, Anh Tuan Luu, Dat Quoc Nguyen

[Who’s Who: Large Language Models Meet Knowledge Conflicts in Practice](#)

EMNLP 2024 (Findings)

Quang Hieu Pham^{*}, Hoang Ngo^{*}, Anh Tuan Luu, Dat Quoc Nguyen

Experience

AI Resident

April 2025 – July 2025

Qualcomm AI Research

- Qualcomm acquired our VinAI’s research team
- Focuses: Agentic AI, Large Language Models, Reasoning, Hallucination

AI Resident

Aug 2023 – April 2025

VinAI Research

- Focuses: Large Language Models, Reasoning, Hallucination

Projects

FlexSQL Agent	University of Alberta
◦ Develop an agentic framework that explores the database on demand, plans multiple interpretations in parallel and execute flexibly in Python or SQL. ◦ Achieve 65.4% pass@1 on Spider2-snow benchmark with gpt-oss-120b, ahead of leaderboard entries built on DeepSeek-R1 and o3.	
Text-to-SQL Translation	VinAI Research
◦ Finetune LLMs on Bird Benchmark. ◦ Propose data augmentation, schema filtering & column suggestion techniques. ◦ Yield 65.58% accuracy on the public dev set with a 9B model. ◦ Best VinAI Resident's applied project Q2 2024.	
E-commerce Agent	Qualcomm AI Research
◦ Design a personal shopping database. ◦ Propose and implement initial prototypes.	

Honors & Awards

Google DeepMind AI Master's Scholarships	Fall 2025
U of A's Department of Computing Science Recruitment Award.	Fall 2025
HUST's scholarship for excellent study results (3.91/4.00 GPA).	Fall 2022
HUST's scholarship for excellent study results (4.00/4.00 GPA).	Spring 2022
Erasmus+ Exchange Scholarship.	Aug 2021
Top 0.1% in Vietnam's national university examination (Math-Physics-English).	Aug 2019

Skills

Programming Languages: Python, C++, C, Java, SQL

Technologies: PyTorch, Langchain, Pandas, Numpy, Git

References

◦ Asst. Prof. Xi Ye 	University of Alberta
◦ <i>MSc. supervisor</i>	
◦ Asst. Prof. Jocelyn Chen 	University of Alberta
◦ <i>MSc. supervisor</i>	
◦ Assoc. Prof. Anh Tuan Luu 	Nanyang Technological University
◦ <i>Head of NTU NAIL</i>	
◦ Dr. Dat Quoc Nguyen 	Qualcomm AI Research
◦ <i>Principal Researcher</i>	
◦ Dr. Tung Pham 	Qualcomm AI Research
◦ <i>Senior Researcher</i>	